$$= \frac{P(X>m+n)}{P(X>n)} = \frac{q^{m+n}}{q^n} = q^m$$

$$= P(X>m)$$

$P(X>m+n \mid X>n) = P(X>m)$

Memoryless property of Geometric distribution.

6. Negative Binomial Distribution

Consider independent Bernoullians trials under identical conditions till rth success is allowed.

$X : r, r+1, r+2, \ldots$

$$P(X=k) = \binom{k-1}{r-1} q^{k-r} p^r, \quad k = r, r+1, r+2, \ldots$$

$$E(X) = \frac{r}{p}, \quad V(X) = \frac{rq}{p^2}$$

$$M_X(t) = E(e^{tX}) = \sum_{k=r}^{\infty} e^{tk} \binom{k-1}{r-1} q^{k-r} p^r$$

$(1-qe^t)^{-r}$

$$= \sum_{k=r}^{\infty} \binom{k-1}{r-1} (qe^t)^{k-r} (pe^t)^r = (pe^t)^r \sum_{k=r}^{\infty} \binom{k-1}{r-1} (qe^t)^{k-r}$$

$$\boxed{M_X(t) = \frac{(pe^t)^r}{(1-qe^t)^r}, \quad qe^t < 1} \text{ or } t < -\log q$$

Ex: Suppose an airplane fails if 2 of its engines fail where the probability of failure of each engine is p.

$$P(X=k) = \binom{k-1}{2-1} q^{k-2} p^2, \quad k = 2, 3, \ldots$$

$= (k-1) q^{k-2} p^2, \quad k = 2, 3, \ldots$

7. Hypergeometric distribution:

Suppose a population consists of N items $\begin{cases} K \text{ - type I} \\ N-K \text{ type II} \end{cases}$

n items are selected at random (without replacement)

X denotes no of items of type-I in the selected sample

$$p_X(x) = P(X=x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, \ldots, n.$$

$$\max(0, n-K+x) \le x \le \min(n, K)$$

$$\sum_{x=0}^{n} \binom{K}{x}\binom{N-K}{n-x} = \binom{N}{n}$$

$$E(X) = K \cdot \frac{n}{N}$$

$$V(X) = \left(\frac{N-n}{N-1}\right) \cdot \frac{Kn}{N}\left(1 - \frac{K}{N}\right)$$

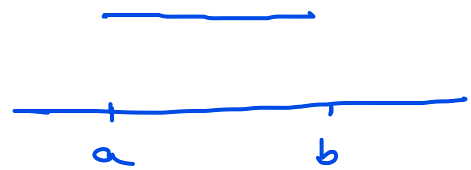
Theorem: Let $X \sim \text{Hypergeo}(K, N, n)$. If $K \to \infty$, $N \to \infty$ such that $\frac{K}{N} \to p$, then $P(X=x) \to \underbrace{\binom{n}{x} p^x (1-p)^{n-x}}_{\text{Binomial distribution } (n,\, p)}, \quad x = 0, 1, \ldots, n$

Special Continuous Distribution:

$$\int_a^b f(x)\,dx = \frac{1}{b-a}\int_a^b dx$$

1. Continuous Uniform Distribution

$$f_X(x) = \frac{1}{b-a}, \quad a \le x \le b$$
$$= 0, \quad \text{otherwise}$$



$$E(X) = \frac{a+b}{2}, \quad V(X) = \frac{(b-a)^2}{12} = \sigma^2$$

$$M_X(t) = E(e^{tX}) = \int_a^b e^{tx} \cdot \frac{1}{b-a}\,dx = \frac{1}{b-a}\left[\frac{e^{tx}}{t}\right]_a^b$$

$$M_X(t) = \frac{e^{tb} - e^{ta}}{t(b-a)}, \quad t \neq 0$$
$$= 1, \quad t = 0$$